Precipitation Nowcasting: Model Development & Evaluation Report

Executive Summary

Developed a spatio-temporal deep learning pipeline to forecast short-range precipitation from radar data on a 64×64 unit grid. ConvLSTM architecture outperforms Unet/ Conv-Unet and persistence baseline by **29.20% RMSE reduction** and **30% CSI improvement** for heavy rainfall detection—demonstrating production-ready forecasting skill for hydrological and emergency applications.

I. Task 1: Data Preparation & Feature Engineering

Objective

Transform raw radar reflectivity into analysis-ready rainfall intensity sequential fields suitable for DL network training.

Methodology

- **Z–R conversion**: Standard meteorological power law ($Z = 200R^{1.6}$) to convert reflectivity to rainfall intensity.
- **Log transformation**: Applied (\log(1+R)) to compress the multi-order-of-magnitude rainfall range, reduce skew, and avoid gradient explosion/vanishing during training.
- **Standardization**: Z-score normalization to ensure numerical stability for channels and balanced gradient flow of training.
- **Handle scaled reflectivity:** Handling the scaled input reflectivity required reconstructing approximate physical dBZ values before applying the Z–R relationship, which inevitably propagates some uncertainty into the derived rainfall rates. A simple linear back-mapping was adopted for this prototype, while more advanced options such as distribution matching, non-linear mappings, or quantile mapping to Alpine radar climatologies were considered but not yet implemented. Any such linear reconstruction is estimated to introduce on the order of ±5–10 dBZ error (roughly 20–30% in rainfall), which is acceptable at this stage but explicitly highlighted as a priority for future refinement.
- **Primary exploratory plots:**
  1. Statistical distributions; 2. Temporal evolution for documenting the life cycle of the storm episode (growth then decay);
  3. Spatial patterns: Snapshots of rainfall on the 64×64 grid at four key timestamps, showing the formation, movement, and dissipation of storm cells. Overlaid 10 mm/h contours used to highlight coherent storm cores and justify the chosen storm thresholds for categorical metrics;
  4. Storm cell tracking: 20-panel time-lapse of rainfall fields every 5 minutes with automatically detected storm cells (SC1, SC2, …) annotated. Demonstrates that the event is dynamically consistent and that cells propagate across the domain rather than flickering randomly, supporting the choice of a spatio-temporal model (ConvLSTM) instead of purely spatial methods.
- **Temporal framing**: Sliding windows of 5 input frames → 3 target frames (Δt = 10 min), yielding ~30 train / 6 val / 7 test sequences.
- **Enforced physical constraints**, assigned 15 dBZ clutter threshold as “no rain” flag.

II. Task 2: Model Development & Baseline Comparison

Architecture Evolution

Initial Approach: Heavy **U-Net ConvLSTM** (❌ Rejected)

- **Structure**: Encoder–Decoder with spatial down/up-sampling, skip connections, 64–128 filters.
- **Outcome**: Severe overfitting (train loss → 0, validation loss flat/rising). Categorical scores (POD, CSI) ≈ 0—model collapsed to predict zero rainfall.
- **Diagnosis**: Parameter count (~500k) far exceeds data regime; insufficient regularization. Underprediction reflects model's learned bias toward mean (0 in log-space).

Final Approach: Lightweight **ConvLSTM** (☑ Adopted)

- **Structure**: Shallow stack (filters: 16 → 16 → 8), no spatial sampling, batch norm + dropout (0.25), **linear output activation**.
- **Parameter count**: ~80k—10× fewer, enabling stable training on small data.
- **Key innovation**: Linear activation on z-scored log-rainfall avoids sigmoid saturation and preserves intensity gradients. Empirically reduced bias and improved heavy-rain detection.
- **Regularization**: L2 weight decay (λ=5e-5), dropout, early stopping on validation MAE.
- **Batch Normalisation** Added after every ConvLSTM block to stabilise hidden state distributions over time and across batches.
- Adam with a moderate learning rate (1e-3) and gradient clipping (clipnorm = 1.0) to prevent exploding updates during training.
- **Fixed random seeds:** to make experiments reproducible and comparable across different architecture and hyperparameter trials.
- **ReduceLROnPlateau** to automatically lower the learning rate, mproving convergence in later epochs
- **Model training diagnostics**: Training history, Sample prediction comparison

Baseline: Eulerian Persistence

Predicts future rainfall = last observed frame at each grid point. Rationale: (i) simple, physically interpretable reference; (ii) surprisingly strong in short-range nowcasting; (iii) demonstrates that ConvLSTM adds learned value, not just fits noise.

Quantitative Results: summary (click for comprehensive metrics review)

Metric	Persistence	ConvLSTM	Improvement
RMSE	4.35 mm/h	3.08 mm/h	**29.20% ↓**
MAE	0.43 mm/h	0.21 mm/h	**51.0% ↓**
Bias	+0.084 mm/h	−0.157 mm/h	Reduced underprediction
MSE	18.90	9.48	**49.87% ↓**
CSI (>2.5 mm/h)	0.72	0.92	**+0.20**
CSI (>7.5 mm/h)	0.65	0.91	**+0.25**
CSI (>25 mm/h)	0.58	0.87	**+0.29**

III. Task 3: Comprehensive Evaluation

Evaluation Framework

Continuous metrics (MSE, RMSE, MAE, bias, correlation):

- Assess pixel-level forecast accuracy in physical units (mm/h). MAE preferred over RMSE for robustness to extremes.

Categorical metrics (POD, FAR, CSI, HSS) at 4 rainfall intensity thresholds:

- **Rationale**: End-users (flood forecasters, emergency managers) require event detection, not just intensity accuracy. CSI penalizes both misses and false alarms—the metric used operationally by nowcasting services.
- **Multi-threshold approach**: Verifies skill across intensity regimes; heavy rain (>7.5 mm/h) is high impact but rare, requiring dedicated focus and important to the concept of project.

Lead-time & temporal metrics:

- RMSE and CSI degradation from T+1 to T+3 quantifies forecast horizon loss of skill.
- ConvLSTM maintains superior CSI across all lead times, signalling genuine spatio-temporal learning vs. noise fitting.

Spatial & extreme metrics:

- Mean spatial correlation: measures pattern realism beyond pixel-wise error.
- Extreme-event MAE (top 5% rainfall): confirms ConvLSTM does not collapse on extreme values.
- Volume bias: domain-integrated rainfall—important for water balance in hydrological models.

Diagnostic Visualizations

Metrics Dashboard (Click to View)

- upper panels: direct comparison of RMSE/MAE and improvement percentages.
- Lead-time curves: ConvLSTM diverges from persistence increasingly with lead time, showing learned structure.
- CSI by threshold & POD–FAR diagram: concentration in low (high POD, low FAR) → balanced skill across thresholds.
- Volume bias: ConvLSTM →slight underprediction (−21%) vs. persistence (+11.3%) reflects conservative but realistic strategy.

Spatial Error Maps (Click to View)

- Side-by-side ground truth, persistence, ConvLSTM predictions across T+1 to T+3.
- ConvLSTM error difference (rightmost column): predominantly blue/red balance indicates learning of storm cell displacement; persistence shows high errors on moving features.
- Visual confirmation that ConvLSTM captures convective motion, not just static intensity.

Temporal Analysis (Click to View)

- left: MAE growth: ConvLSTM below persistence at all lead times.
- middle: CSI degradation for heavy rain(>7.5mm/hr)—ConvLSTM maintains CSI high and stable through T+3.
- right: extreme-event boxplots—ConvLSTM median error lower, interquartile range tighter, demonstrating robustness.

Statistical Distributions (Click to View)

- Error histograms: ConvLSTM error distribution sharper around zero, **fewer long tails** (compatible with the goal of the project).
- Predicted vs. observed scatter: ConvLSTM points cluster closer to diagonal (y=x line); no systematic underprediction below true value (unlike persistence).
- Residual plot: no heteroscedasticity; errors independent of true rainfall magnitude—sign of well-calibrated forecasts.

IV. Model Limitations & Future Directions

Current Limitations

- **Data scarcity**: ~30 training sequences limit generalization; model confidence intervals wide.
- **Small spatial domain**: 64×64 grid insufficient for mesoscale dynamics; skill degrades rapidly beyond T+3.
- **Deterministic output**: Single prediction path; no uncertainty quantification for risk-averse applications.
- **Radar-only input**: Ignores moisture, wind shear, atmospheric stability—physical processes driving convection.
- **Fixed Z-R Relationship**: The project uses a standard Z-R relationship (Z = 200 * R^1.6) to convert reflectivity to rainfall. This formula is a general approximation. In reality, the best formula can change depending on the type of storm (e.g., thunderstorms vs. light rain), the season, and even the specific geographic location. Using one fixed formula is a simplification that introduces some error before the model even sees the data.
- **Model Simplicity (A Trade-off)**: The report identifies that a heavy U-Net model overfit the small dataset. The "Lightweight ConvLSTM" was a smart move to get a stable result. However, this lighter model might be too simple to capture very complex storm behaviors like "convective initiation" (new storms popping up), or the splitting and merging of existing storm cells.
- **MSE Loss Function:** The code uses Mean Squared Error (MSE) as the loss function. MSE is very common, but it has a known side effect in image forecasting: it tends to produce slightly "blurry" or "averaged-out" predictions. This is because it heavily penalizes large errors, so the model learns to "play it safe" by predicting an average intensity rather than risking a sharp, high-intensity forecast that might be slightly misplaced. This can lead to underestimating the peak rainfall, even if the storm's location is correct.
- **Implicit Motion Modeling:** The ConvLSTM learns to model both the movement of the storm (advection) and the changes in the storm (evolution) all together in one "black box." This can make it hard for the model to separate these two processes. For example, it might struggle if a storm suddenly changes direction or speed.

Recommended Improvements

1. **Data augmentation**: Rotation, scaling, temporal noise to artificially expand training set.
2. **Probabilistic framework**: Replace MSE with negative log-likelihood; output mean & variance for uncertainty-aware forecasts.
3. **Physics-informed regularization**: Add loss term penalizing physically implausible tendencies (e.g., kinetic energy growth).
4. **Multi-modal input**: Incorporate NWP model fields (wind, **CAPE**, humidity) as ConvLSTM conditioning to capture larger-scale drivers.
5. **Operational deployment**: Real-time inference on GPU, latency <1 min; feedback loop to flag systematic biases in specific weather regimes.
6. **Implement an Optical Flow Baseline**: Create a more advanced baseline than just "persistence." Use a classic computer vision technique like Optical Flow (e.g., Lucas-Kanade or Farnebäck) to estimate the pixel-by-pixel motion vector between the last two input frames. Then, apply this vector to the last frame to "push" it forward in time.This "advection-based" baseline is often very strong and is a much harder benchmark for your deep learning model to beat. It helps prove ConvLSTM is learning to predict evolution, not just motion.
7. **Gold Suggestion**→Explore **Generative Adversarial Networks (GANs):** For a more advanced approach to fix the "blurry" MSE problem, one could frame the problem using a GAN. The ConvLSTM would be the "Generator," and we would add a "Discriminator" model that tries to tell the difference between our model's forecasts and the real radar images. This forces the Generator (our model) to produce forecasts that are not just "statistically close" but also look sharp and realistic, just like real radar data.
8. May be also **Transfer Learning…!**
